Unit A4: Key Concepts Cheat Sheet

ML Types

- **Supervised Learning:** Learning with a teacher. Data has input X and correct label Y. (e.g., Classification, Regression).
- **Unsupervised Learning:** Learning without a teacher. Data has input X only. Model finds structure. (e.g., Clustering).
- **Reinforcement Learning:** Learning by trial and error. Agent interacts with environment to maximize Reward.
- **Deep Learning:** Using multi-layered Neural Networks to learn from vast amounts of data (e.g., Image/Speech recognition).

Data Issues & Preprocessing

- **Outliers:** Data points significantly different from others (can skew models).
- **Imputation:** Filling in missing values (e.g., with Mean/Median).
- **Normalization:** Scaling features to a standard range (0-1) to ensure fair comparison.
- **Feature Selection:** Choosing only relevant variables to improve model performance.
- **Overfitting:** Model learns noise/details of training data, performs poorly on new data. (High Variance).
- **Underfitting:** Model is too simple to capture the underlying trend. (High Bias).

Evaluation Metrics (Confusion Matrix)

- **True Positive (TP):** Correctly predicted Yes.
- **True Negative (TN):** Correctly predicted No.
- **False Positive (FP):** Predicted Yes, but actually No (Type I Error).
- **False Negative (FN):** Predicted No, but actually Yes (Type II Error).
- **Accuracy:** $(TP + TN)/Total$. Overall correctness.
- **Precision:** $TP/(TP + FP)$. Accuracy of positive predictions.
- **Recall:** $TP/(TP + FN)$. Coverage of actual positives.
- **F1 Score:** Harmonic mean of Precision and Recall.

Neural Network Terms

- **Neuron (Perceptron):** Basic processing unit. $Output = Activation(Weight \times\ Input + Bias)$.
- **Hidden Layer:** Layers between Input and Output where feature extraction occurs.
- **Activation Function:** Decides if a neuron "fires" (e.g., Sigmoid, ReLU). Adds non-linearity.

- **Convolutional Layer:** (CNN) Applies filters to input to create feature maps (detects edges).
- **Pooling Layer:** (CNN) Reduces dimensionality (e.g., Max Pooling takes the largest number in a grid).

Genetic Algorithm Steps

1. **Initialization:** Create random population.
2. **Evaluation:** Calculate fitness of each individual.
3. **Selection:** Choose parents based on fitness.
4. **Crossover:** Combine parents to create offspring.
5. **Mutation:** Randomly change genes to maintain diversity.
6. **Termination:** Stop when goal is met or generations run out.